A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions

Xiaoning Bian[*] Sarah Meng Li[†‡] Neil J. Ross[§] John van de Wetering[†‡] Yuming Zhao[¶]

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[*]Tsinghua University [†]University of Amsterdam [‡]QuSoft [§]Dalhousie University
[¶]University of Copenhagen

This talk

- A presentation by generators and relations of the n-qudit Clifford group, for every odd prime dimension p: 16 relations, none involving more than 3 qudits.
- A unique normal form for symplectic (= Clifford modulo Pauli) circuits.
- All, except for two math facts, are formalised in Agda.

About this slides

- Every Agda code block is typechecked by Agda.
- Written by *Claude* and Xiaoning Bian ($\sim$200-line outline prompt and slides checking & improvement).

Introduction

From qubit to qudit Clifford gates

Fix an odd prime p and let $\omega = e^{2\pi i/p}$. The Clifford group $\mathcal{C}_n$ is generated by:

	qubit ($p = 2$)	qudit (odd prime p)
H	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$\lvert j\rangle \mapsto \frac{1}{\sqrt{p}} \sum_k \omega^{jk} \lvert k\rangle$
S	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$\lvert j\rangle \mapsto \omega^{j(j-1)/2} \lvert j\rangle$
CZ	$\mathbf{diag}(1,1,1,-1)$	$\lvert j\rangle\lvert l\rangle \mapsto \omega^{jl} \lvert j\rangle\lvert l\rangle$

Note the $/2$ in the qudit S is the inverse of 2 in $\mathbb{Z}_p$. A global phase can be added to the defition of H to facilitate computation.

The setting: one Agda module

These slides are a single Agda module, parameterised by the prime p and by a primitive root g of $\mathbb{Z}_p^*$:

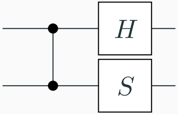
```
module qpl26slides.talk
  (p-3 : ℕ)
  (let p-2 = 1+ p-3)
  (p-prime : Prime (3+ p-3))
  (let open PrimeModulus' p-2 p-prime)
  (g*@(g , g≠0) : ℤ* ₚ)
  (g-gen : ∀ ((x , _) : ℤ* ₚ) → ∃ \ (k : ℤ ₚ₋₁) → x ≡ g ^′ toℕ k)
  where
```

- $p =$ `p-3` $+ 3$: every odd prime, uniformly.
- Everything downstream — rule sets, normal forms, theorems — is parameterised by the choice of g.

Circuits are words

A gate set is a family of gate types indexed by arity; a circuit on n wires is a word over wire-indexed generators.

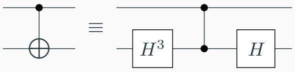
```
data SympGate : ℕ → Set where
  H-gate  : SympGate 1
  S-gate  : SympGate 1
  CZ-gate : SympGate 2

example : Circuit 2
example = CZ • H ↑ • S
```

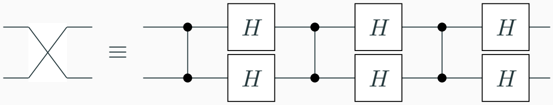
- `Circuit` n $=$ `Word` (`Gen` n): free monoid; `_↑` shifts a circuit up one wire.
- diagrams read in matrix-multiplication order: `example` $=$ 

Derived gates: the rest of the usual gate set

X, Z, the swap Ex, and the two controlled-X gates are *words*, not extra generators:

```
Z X : ∀ {n} → Word (Gen (1+ n))
Z = H • H • S • H • H • S⁻¹
X = H • S • H • H • S⁻¹ • H

Ex CX XC : ∀ {n} → Word (Gen (2+ n))
Ex = CZ • H ↓ • H ↑ • CZ • H ↓ • H ↑ • CZ • H ↓ • H ↑
CX = H ↓ ^ 3 • CZ • H ↓
XC = H ↑ ^ 3 • CZ • H ↑
```





Main results

Main result 1: a presentation of the qudit Clifford group

The paper's Figure 1 rule set, mod scalars, presents the projective Clifford group $\mathcal{C}_n/\langle\omega\rangle \;\cong\; \mathrm{Pauli}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p)$:

```
Theorem-1 : ∀ (n : ℕ) → (n CRel,_===_) IsPresentationOf (Pauli⋊Sp n)
Theorem-1 = PapPres.presentation
```

- `_CRel,_===_` is the 16-relation rule set, shown on the next slides.
- The theorem holds *at every width* n, for *every* odd prime p and every primitive root g — the whole deck is parameterised by them.

The 16 relations, in Agda (one qudit)

```
data _CRel,_===_ : (n : ℕ) → WRel (Gen n) where
  order-S     :       (₁₊ n) CRel, S ^ p === ε
  order-H     :       (₁₊ n) CRel, H ^ 2 === M₋₁
  M-power     : ∀ k → (₁₊ n) CRel, Mg^ k === M (g^ k)
  semi-MR     :       (₁₊ n) CRel, R • Mg === Mg • R^ (g * g)
  order-SH    :       (₁₊ n) CRel, (S • H) ^ 3 === ε
  comm-HHSHHS :       (₁₊ n) CRel, H • H • S • H • H • S === S • H • H • S • H • H
```

- M_x is the multiplier $|j\rangle \mapsto |xj\rangle$, spelled as the word $R^{x^{-1}}\, H\, R^x\, H\, R^{x^{-1}}\, H$ with $R = S \cdot Z^{1/2}$.
- One-qudit relations talk about the powers of a fixed primitive root g — this is where odd primes differ from the qubit case.

The 16 relations, in Agda (two qudits)

```
order-CZ   : (₂₊ n) CRel, CZ ^ p === ε
order-Ex   : (₂₊ n) CRel, Ex ^ 2 === ε
comm-CZ-S↑ : (₂₊ n) CRel, CZ • S ↑ === S ↑ • CZ
semi-M↑CZ  : (₂₊ n) CRel, CZ • Mg ↑ === Mg ↑ • CZ^ g
semi-Ex-S↑ : (₂₊ n) CRel, Ex • S ↑ === S ↓ • Ex
semi-Ex-H↑ : (₂₊ n) CRel, Ex • H ↑ === H ↓ • Ex
blake-c12  : (₂₊ n) CRel, (S ^ p-1) ↑ • (S ^ p-1) ↓ • CX ^ p-1 • S ↓ • CX === CZ
```

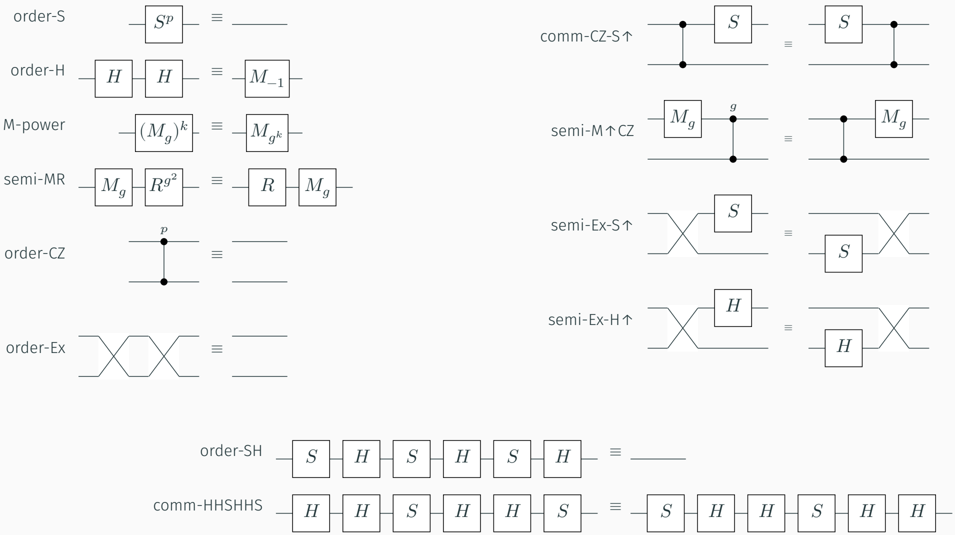
- `blake-c12` converts between the two entangling gates CZ and CX.

The 16 relations, in Agda (three qudits)

```
yang-baxter : (₃₊ n) CRel, Ex ↑ • Ex ↓ • Ex ↑ === Ex ↓ • Ex ↑ • Ex ↓
cz-slide    : (₃₊ n) CRel, Ex ↓ • Ex ↑ • CZ === CZ ↑ • Ex ↓ • Ex ↑
semi-CX↑-CZ↓ : (₃₊ n) CRel, CZ ↓ • CX ↑ === CZ02 • CX ↑ • CZ ↓
```

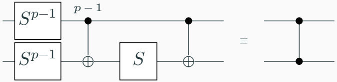
- Only these 16 are axioms. $XZ = ZX \,(\mathrm{mod}\ \omega)$, $\boldsymbol{Ex}$-vs-CZ, and both Pauli-vs-CZ rules are *derived*.
- Plus, for every gate set, the same three monoidal structural rules: congruence under $\uparrow$, gates on disjoint wires commute, and scalars commutes with every gate.

The 16 relations, as circuits (1–2 qudits)

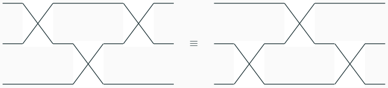


The 16 relations, as circuits (2–3 qudits)

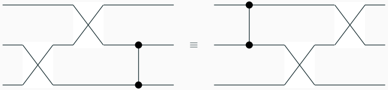
blake-c12



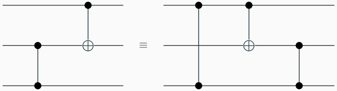
yang-baxter



cz-slide



semi-CX↑-CZ↓



What “presents” means

```
record _Presents_ (_===_ : WRel X) (G : Group a ℓ) : Set (a ⊔ ℓ) where
  field gl : Grouplike _===_
  module GL = Group-Lemmas _===_ gl
  open GroupMorphisms (Group.rawGroup GL.•-ε-group) (Group.rawGroup G)
  field
    ⟦_⟧ : Word X → Group.Carrier G
    iso : IsGroupIsomorphism ⟦_⟧
```

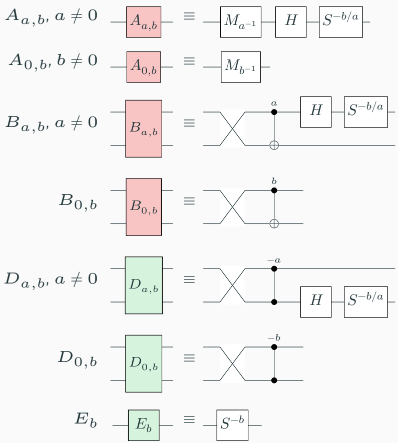
$$\mathsf{Cir}\ n \,/\, \approx_{\text{16 rules}} \xrightarrow[\cong]{⟦\_⟧} \mathrm{Pauli}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p) \;\cong\; \mathcal{C}_n/\langle\omega\rangle$$

- *Soundness*: ⟦_⟧ is well defined — each rule holds semantically.
- *Completeness*: ⟦_⟧ is injective — semantically equal circuits are related by the rules.

Main result 2: a unique normal form for symplectic operators

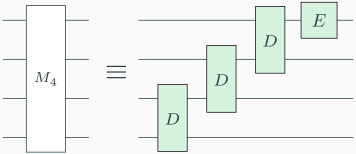
Symplectic operators are Clifford mod Pauli: the “proper” part. Normal forms are data, built from four atomic boxes:

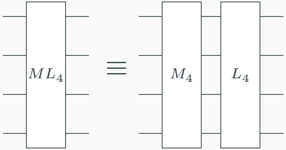
```
A B D E : Set
A = Σ[ ab ∈ (ℤ p × ℤ p) ]
  (ab ≢ (0 , 0))
B = ℤ p × ℤ p
D = ℤ p × ℤ p
E = ℤ p
```

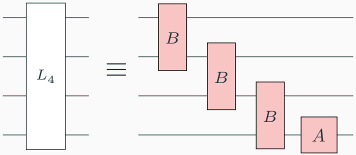


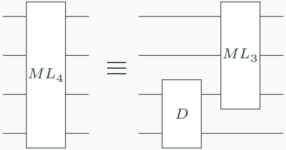
Rows, and the staircase

```
M L ML : ℕ → Set

M 0 = ⊤

M (1+ n) = Vec D n × E

L 0 = ⊤

L (1+ n) = Vec B n × A

ML 0 = ⊤

ML 1 = M 1 × L 1

ML (2+ n) =

  M n × L n ⊎ D × ML (1+ n)
```





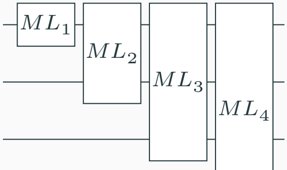




The normal form, as a circuit

Reading normal forms back as circuits is structural recursion:

```
NF : ℕ → Set
NF 0 = ⊤
NF (1+ n) = NF n × ML (1+ n)


[_]ⁿᶠ : ∀ {n} → NF n → Word (Gen n)
[_]ⁿᶠ {0}    tt         = ε
[_]ⁿᶠ {1+ n} (nf , ml) = [ nf ]ⁿᶠ ↑ • [ ml ]ᵐˡ
```



Everything on these slides is Agda-checked

- ≈ 320 files, $\approx 100\,000$ lines of Agda, all `--safe`, no postulates.
- Agda 2.8, standard library 2.3.
- One root (`MainTheorems.agda`) reaches the whole library; these slides import the library and *re-state* the main theorems — if a statement on a slide did not match the formalisation, the slides would not build.

What the formalisation covers
All group-theoretic computation: presentations, normal forms, completeness. The two glue facts — that the projective Clifford group is $\mathbf{Pauli} \rtimes \mathbf{Sp}$, and that the Clifford group is a central extension of it by $\langle\omega\rangle$ — are classical and are *not* formalised.

Method

Divide and conquer

Factor the group; present the factors; compose the presentations.

$$1 \to \mathbf{Pauli}_n \to \mathcal{C}_n/\langle\omega\rangle \to \mathbf{Sp}(2n, \mathbb{Z}_p) \to 1$$

semidirect: it splits

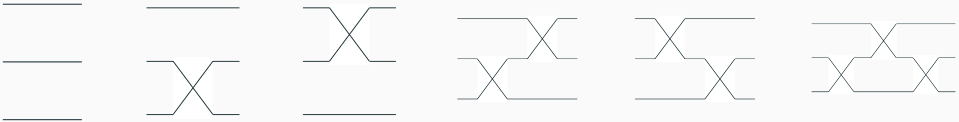
$$1 \to \langle\omega\rangle \to \mathcal{C}_n \to \mathcal{C}_n/\langle\omega\rangle \to 1$$

central extension

- Doesn't this just move the difficulty around? No: the symplectic factor is where the real rewriting happens, and keeping the Pauli part out of it pays off everywhere.
- Prior work (qubit: Selinger; qutrit) worked with the whole Clifford group at once — the derivations drag a "messy" Pauli correction through every step.

Warm-up: the symmetric group S_n

Swap circuits on n wires present S_n. All of S_3:



e $\quad$ σ_0 $\quad$ σ_1 $\quad$ $\sigma_0\sigma_1$ $\quad$ $\sigma_1\sigma_0$ $\quad$ $\sigma_0\sigma_1\sigma_0$

Circuits have a built-in tower of subgroups

$$\mathsf{Cir}\,0 \;<\; \mathsf{Cir}\,1 \;<\; \cdots \;<\; \mathsf{Cir}\,n$$

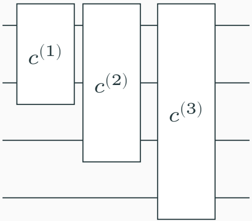
($\mathsf{Cir}\,k$ = circuits that only touch the top k wires). For S_n: $S_0 < S_1 < \cdots < S_n$.

Coset representatives, and the normal form

Representatives of S_3/S_2 (and of S_2/S_1): the staircases

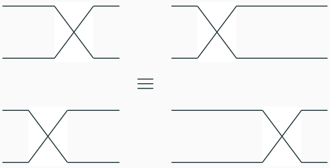
$c_0^{(2)}$ $c_1^{(2)}$ $c_2^{(2)}$ $c_0^{(1)}$ $c_1^{(1)}$

- For $f \in S_3$ there is a unique $c \in C_2$ with $f^{-1}c$ fixing the bottom wire: c must send $f^{-1}(0)$ to 0.
- Recurse on $f^{-1}c \in S_2$: every f is uniquely $c^{(2)} \cdot (c^{(1)}\uparrow)$.
- Group theory: given $G_0 < \cdots < G_n$ and coset reps C_k of G_k/G_{k-1}, elements of G_n factor uniquely as $\prod_k c_k$ — a mixed-radix encoding: $|S_3| = 3 \cdot 2 \cdot 1$.

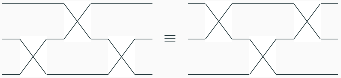


Rewriting to the normal form: push a generator through

Multiply a normal form by a generator; push it through the staircase:



disjoint wires: commute past



Yang–Baxter: pass through, shift up

$\sigma^2 = e$: staircase shrinks

From S_n to $\mathrm{Sp}(2n, \mathbb{Z}_p)$: change the action

The same framework, three ways of looking at the wires:

1. S_n acts on $\{0, \ldots, n-1\}$: a wire is a *position*; c sends $f^{-1}(0)$ to 0.
2. S_n acts on **Vec Bit** n: a wire is a *bit*; $f^{-1}c$ fixes the linear subspace of the bottom wire.
3. Adding H, S, CZ: the group acts on **Vec Pauli**$_1$ n — a $2n$-dimensional symplectic $\mathbb{Z}_p$-vector space. A wire is a $\mathbf{Pauli}_1$; enlarging the coset layer from staircases to $\boldsymbol{ML}$ boxes, $f^{-1}c$ fixes the symplectic subspace of the bottom wire.

The framework applies to *any* $\mathbb{N}$-indexed family of groups with such a layered structure (`Circuit.CosetNF`, `Circuit.Uniqueness`).

Uniqueness, in Agda

```
Theorem-2 : ∀ n {u v : NF n} → ⟦ [ u ] ⟧ ≈ˢ ⟦ [ v ] ⟧ → u ≡ v

Theorem-2 = U.⟦[]⟧-injective
```

$$\begin{tikzcd}
\mathsf{Cir}\, n \,/\approx \arrow[rr, "⟦_⟧"] \arrow[d, bend left, "nf"] & & \mathrm{Sp}(2n, \mathbb{Z}_p) \\
\mathit{NF}\ n \arrow[u, bend left, "[_]"] \arrow[urr, dashed, "\text{injective}"'] & &
\end{tikzcd}$$

Soundness + unique normal forms ⇒ completeness (**`by-normalization`**); the interesting content is that the *coset factor is determined by the action* on Pauli states.

From 66 relations to 18

1. Rewrite *every* circuit to normal form; collect every relation the rewrite ever uses: 66 relations (box-vs-gate commutations, case by case).
2. By inspection, find a small set of primitive, bidirectional relations that derive all 66.
3. The choice is not canonical — pick the set you find natural.

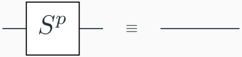
An example of a derived rule, $M_x \cdot M_y = M_{xy}$:

$$-\boxed{M_x}-\boxed{M_y}- \;\equiv\; -\boxed{M_{xy}}-$$

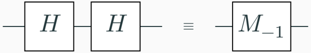
```
M-mul : ∀ n (x y : ℤ* ₚ) → let open PB (Sim._QRel,_===_ (₁₊ n)) in
        M x • M y ≈ M (x *' y)
M-mul n = SimL.Lemmas1.lemma-M-mul n
```

($x = g^k$, $y = g^l$; `M-power` twice; add exponents mod $p - 1$.)

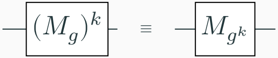
order-S



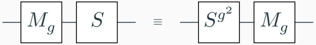
order-H



M-power

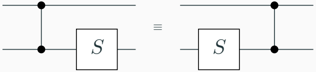


semi-MS



order-CZ

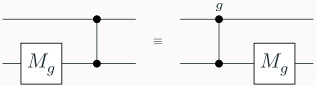
comm-CZ-S↓



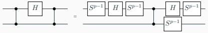
comm-CZ-S↑

semi-M↑CZ

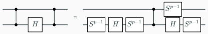
semi-M↓CZ



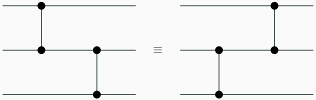
selinger-c10



selinger-c11

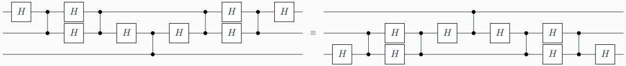


selinger-c12

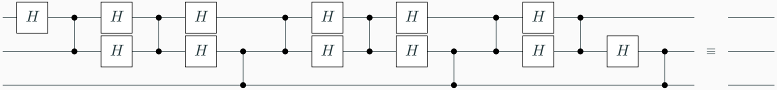


The simplified symplectic rule set (3 qudits), and the theorem

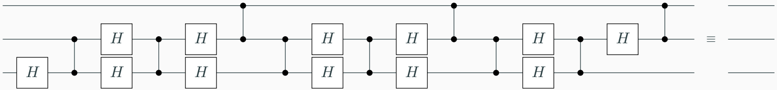
selinger-c13



selinger-c14



selinger-c15

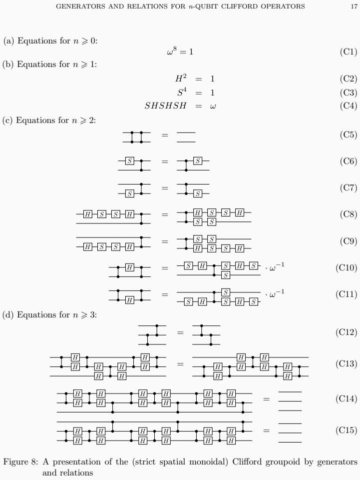


```
Theorem-Sp : ∀ {n} → (n SRel,_===_) IsPresentationOf (Sp-group n)
Theorem-Sp = SimPres.presentation
```

Comparison with the qubit rule set (Selinger)

- identical 3-qudit relations (C12–C15);
- almost identical 2-qudit relations;
- more complicated 1-qudit relations — the odd-prime multiplier calculus (M_x, primitive root g) has no qubit counterpart.

Figure 8 of Selinger, *Generators and relations for n-qubit Clifford operators* →


GENERATORS AND RELATIONS FOR n-QUBIT CLIFFORD OPERATORS 17

(a) Equations for $n \geqslant 0$:

$$\omega^8 = 1 \quad \text{(C1)}$$

(b) Equations for $n \geqslant 1$:

$$H^2 = 1 \quad \text{(C2)}$$
$$S^4 = 1 \quad \text{(C3)}$$
$$SHSHSH = \omega \quad \text{(C4)}$$

(c) Equations for $n \geqslant 2$:

(d) Equations for $n \geqslant 3$:

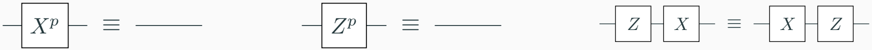
Figure 8: A presentation of the (strict spatial monoidal) Clifford groupoid by generators and relations

Composition

Composing: the Pauli factor

The projective Pauli group $(\mathbb{Z}_p \times \mathbb{Z}_p)^n$ has an easy presentation over generators X, Z on each wire:

```
data _PRel,_===_ : (n : ℕ) → WRel (Gen n) where
  order-X  : (₁₊ n) PRel, X ^ p === ε
  order-Z  : (₁₊ n) PRel, Z ^ p === ε
  comm-Z-X : (₁₊ n) PRel, Z • X === X • Z

Theorem-Pauli : ∀ {n} → (XZ._QRel,_===_ n) IsPresentationOf (+ₚ-group n)
Theorem-Pauli = XZ.presentation
```



Composing: the semidirect product

Semidirect product: relations of both factors, plus twisted commuting, $h\,n' = ({}^{h}n')\,h$, so $(n,h)\cdot(n',h') = (n \cdot {}^{h}n'\ ,\ h\,h')$:

```
conj : Sym.Gen n → XZ.Gen n → Word (XZ.Gen n)
conj Sym.H-gen  XZ.X-gen        = XZ.Z
conj Sym.H-gen  XZ.Z-gen        = XZ.X ^ p-1
conj Sym.S-gen  XZ.X-gen        = XZ.X • XZ.Z
conj Sym.S-gen  XZ.Z-gen        = XZ.Z
conj Sym.CZ-gen XZ.X-gen        = XZ.X • XZ.Z XZ.↑
conj Sym.CZ-gen (XZ.X-gen XZ.↥) = XZ.X XZ.↑ • XZ.Z

Pauli⋊Symplectic : (n : ℕ) → WRel (XZ.Gen n ⊎ Sym.Gen n)
Pauli⋊Symplectic n = XZ._QRel,_===_ n ⋄ Sim._QRel,_===_ n ⋄ ConjRelʷ conj
```

The projection to the symplectic factor is a homomorphism, so symplectic circuits and their relations lift unchanged.

Plumbing: presentation simplification

Two presentations are equivalent when the presented groups are isomorphic over the same interpretation — each side's axioms derived from the other's. All Agda-checked, e.g. `Paper-V0.Iso`, `Simplified.Iso`.

	`SemiDirect`	`Paper-V0`
generators	X, Z, H, S, CZ per wire	H, S, CZ per wire
relations	Pauli + symplectic + conjugation	16

generator reduction: X, Z become the derived words from the introduction

relation reduction: conjugation rules become derivable, e.g. $CZ \cdot X{\uparrow} \approx X{\uparrow} \cdot Z{\downarrow} \cdot CZ$

Getting the scalars back

The Clifford group itself is a central extension by $\langle\omega\rangle$. On presentations: add a generator ω with $\omega^p = \varepsilon$, make it commute with every gate, and correct each mod-scalar relation by the scalar it actually picks up — only `order-SH` needs one:

$$-\boxed{S}-\boxed{H}-\boxed{S}-\boxed{H}-\boxed{S}-\boxed{H}- \;\equiv\; \boxed{\omega^{(p^2-1)/8}\, I}$$

```
Theorem-Clifford : ∀ (n : ℕ) →
  (QS._Exact,_===_ n) IsPresentationOf
  (PE.Presented-group n (CQP.exact-generator-data n))
Theorem-Clifford = CQP.presentation-exact
```

(δ_p : the exponent $(p^2 - 1)/8$ is exact division — p odd — and the correction is checked, not assumed.)

Conclusion

Conclusion

- Main result 1: presentations by generators and relations of the qudit Clifford operators (16 relations, ≤ 3 qudits) and of the symplectic operators, for all odd primes at once.
- Main result 2: a unique normal form for qudit Clifford and symplectic operators.
- Both are checked in Agda — except for the two classical glue facts: that projective Clifford is $\mathbf{Pauli} \rtimes \mathbf{Sp}$, and that Clifford is a central extension of projective Clifford by the scalars.

Thank you!